

ADVANCED LINEAR ALGEBRA

HOMEWORK 1

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P.A.1 Show that:

- (a) $(1 + i)^4 = -4$
- (b) $(1 - i)^{-1} - (1 + i)^{-1} = i$
- (c) $(i - 1)^{-4} = -1/4$
- (d) $10(1 + 3i)^{-1} = 1 - 3i$
- (e) $(\sqrt{3} + i)^3 = 8i$

Solution. (a) The solution follows by computing $(1 + i)^2$, which follows as

$$\begin{aligned}(1 + i)(1 + i) &= 1 + 2i + i^2 \\ &= 2i\end{aligned}$$

so we see that

$$\begin{aligned}(1 + i)^2(1 + i)^2 &= (2i)^2 \\ &= 4(-1) \\ &= -4\end{aligned}$$

(b) For the problem $(1 - i)^{-1} - (1 + i)^{-1} = i$ set $a = 1 - i$ and $b = 1 + i$. Then notice that

$$\begin{aligned}a^{-1} &= \frac{1}{1^2 + (-1)^2} - \frac{-1}{1^2 + (-1)^2} \\ &= \frac{1}{2} + \frac{1}{2}i\end{aligned}$$

and

$$\begin{aligned}b^{-1} &= \frac{1}{1^2 + 1^2} - \frac{-1}{1^2 + 1^2} \\ &= \frac{1}{2} - \frac{1}{2}i\end{aligned}$$

so

$$\begin{aligned}(1-i)^{-1} - (1+i)^{-1} &= \frac{1}{2} + \frac{1}{2}i - \frac{1}{2} + \frac{1}{2}i \\ &= i\end{aligned}$$

(c) Set $a = i - 1$, then

$$\begin{aligned}a^2 &= (i-1)^2 \\ &= -1 - 2i + 1 \\ &= -2i\end{aligned}$$

So we see that

$$\begin{aligned}a^4 &= -4 \\ a^{-4} &= -\frac{1}{4} \\ (i-1)^{-4} &= -\frac{1}{4}.\end{aligned}$$

(d) Let $a = 1 + 3i$, and notice that

$$\begin{aligned}a^{-1} &= \frac{1}{1^2 + 3^2} - \frac{3}{1^2 + 3^2}i \\ &= \frac{1}{10} - \frac{3}{10}i\end{aligned}$$

Now we have that

$$\begin{aligned}10(1+3i)^{-1} &= 10a^{-1} \\ &= 10\left(\frac{1}{10} - \frac{3}{10}i\right) \\ &= 1 - 3i\end{aligned}$$

(e) We want to show that $(\sqrt{3} + i)^3 = 8i$ so we set $a = \sqrt{3} + i$ and we compute the following

$$\begin{aligned}a^2 &= (\sqrt{3} + i)^2 \\ &= 3 - 1 + 2\sqrt{3}i \\ &= 2 + 2\sqrt{3}i\end{aligned}$$

Then we see that

$$\begin{aligned}a^3 &= (\sqrt{3} + i)(2 + 2\sqrt{3}i) \\ &= 2\sqrt{3} + 2i + 6i - 2\sqrt{3} \\ (\sqrt{3} + i)^3 &= 8i.\end{aligned}$$

□

P.A.3 Let $z = 3 + 4i$. Plot and label the following points in the complex plane: z , $-z$, $\bar{z}$, $-\bar{z}$, $1/z$, $1/\bar{z}$, $-1/\bar{z}$

Solution. We easily see that:

$$\begin{aligned} z &= 3 + 4i \\ -z &= -3 - 4i \\ \bar{z} &= 3 - 4i \\ -\bar{z} &= -3 + 4i \\ \frac{1}{z} &= z^{-1} = \frac{3}{3^2 + 4^2} - \frac{4}{3^2 + 4^2}i = \frac{3}{25} - \frac{4}{25}i \\ \frac{1}{\bar{z}} &= \frac{3}{25} + \frac{4}{25}i \\ -\frac{1}{\bar{z}} &= -\frac{3}{25} - \frac{4}{25}i \end{aligned}$$

So we have our plot given in Figure 1,

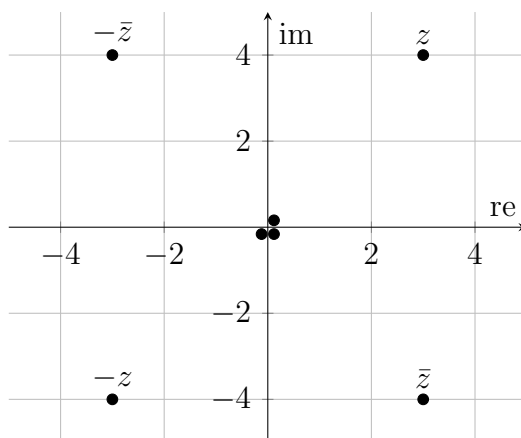


FIGURE 1. Plot of the points z , $-z$, $\bar{z}$, $-\bar{z}$, $1/z$, $1/\bar{z}$, $-1/\bar{z}$

Note that the unlabeled dots around the origin are starting from the top right and moving clockwise around the origin are $1/z$, $1/\bar{z}$, $-1/\bar{z}$. □

P.A.6 Write the following in polar form:

- (a) $1 + i$
- (b) $(1 + i)^2$
- (c) $(1 + i)^4$

Solution. Note that the polar form of z^n as $z^n = r^n(\cos(n\theta) + i\sin(n\theta))$ for $n \in \mathbb{Z}$. (a) For $1 + i$ we get $r = \sqrt{1+1}$ and $\theta = \frac{\pi}{4}$. So,

$$1 + i = \sqrt{2} \left(\cos\left(\frac{\pi}{4}\right) + i \sin\left(\frac{\pi}{4}\right) \right)$$

(b) Then using our formula we have

$$(1 + i)^2 = 2 \left(\cos\left(\frac{\pi}{2}\right) + i \sin\left(\frac{\pi}{2}\right) \right)$$

(c) Again using our formula we have

$$(1 + i)^4 = 8 (\cos(\pi) + i \sin(\pi))$$

□

P.A.7 What are the square roots of $1 + i$? Draw a picture of their locations.

Solution. Using rectangular form we have that for $z = a + bi$

$$\sqrt{a + bi} = \pm \left(\sqrt{\frac{|z| + a}{2}} + i \cdot \frac{b}{|b|} \sqrt{\frac{|z| - a}{2}} \right)$$

So we get that

$$\begin{aligned} \sqrt{1 + i} &= \pm \left(\sqrt{\frac{\sqrt{2} + 1}{2}} + i \sqrt{\frac{\sqrt{2} - 1}{2}} \right) \\ &\approx \pm (1.0987 + 0.4551i) \end{aligned}$$

□

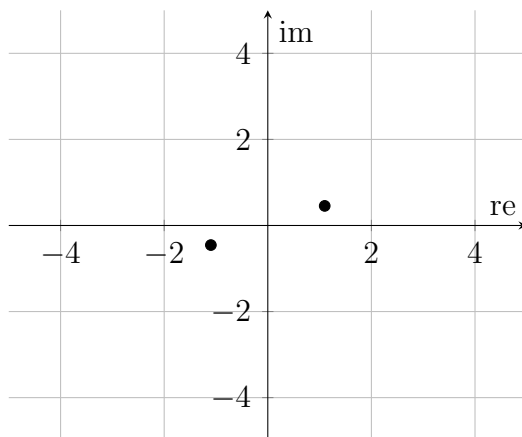


FIGURE 2. Plot of the points for $\sqrt{1 + i}$

P.A.11 If $z \neq 0$, show that $\overline{1/z} = 1/\bar{z}$.

Proof. Let $z = a + bi$, then we have

$$\begin{aligned}\overline{\left(\frac{1}{z}\right)} &= \overline{\left(\frac{a}{a^2 + b^2} - i\frac{b}{a^2 + b^2}\right)} \\ &= \frac{a}{a^2 + b^2} + \frac{b}{a^2 + b^2}i\end{aligned}$$

and similarly

$$\frac{1}{\bar{z}} = (a - bi)^{-1} = \frac{a}{a^2 + b^2} + \frac{b}{a^2 + b^2}i$$

So $\overline{1/z} = 1/\bar{z}$. □

P.B.2 If p is a real polynomial, show that $p(\lambda) = 0$ if and only if $p(\bar{\lambda}) = 0$.

Proof. ($\Rightarrow$) Let $p(\lambda) = 0$. So there must be some polynomial $q(z)$ such that $p(z) = (z - \lambda)q(z) + r$. Also notice that

$$\begin{aligned}p(\lambda) &= (\lambda - \lambda)q(\lambda) + r = 0 \\ &= r = 0.\end{aligned}$$

Now set $\lambda = a + bi$ so that $\bar{\lambda} = a - bi$. Now what is $p(\bar{\lambda})$? Well its

$$p(\bar{\lambda}) = (a - bi - a + bi)q(\bar{\lambda}) = 0.$$

($\Leftarrow$) Similarly to the previous case let $p(\bar{\lambda}) = 0$ and we find a polynomial $q(z)$ such that $p(z) = (z - \bar{\lambda})q(z)$ (shortcutting the dropping of the constant term). Then we see that

$$p(\lambda) = (\lambda - \bar{\lambda})q(\lambda) = (a + bi - a - bi)q(\lambda) = 0$$

□

P.C.3 If $A \in \mathbf{M}_n(\mathbb{F})$ is invertible, show that $A^\top$ is invertible and $(A^{-1})^\top$ is its inverse.

Proof. We have the standard identities $(XY)^\top = X^\top Y^\top$ and $I^\top = I$. Now let $A \in \mathbf{M}_n(\mathbb{F})$ be invertible so then we have that $AA^{-1} = I$ and we have that $A^{-1}A = I$. So let's apply our standard identities to these equations, and we have

$$(AA^{-1})^\top = I^\top \quad \Rightarrow \quad (A^{-1})^\top A^\top = I$$

and

$$(A^{-1}A)^\top = I^\top \quad \Rightarrow \quad A^\top (A^{-1})^\top = I$$

So $(A^{-1})^\top$ is the left and right inverse of $A^\top$. □

P.C.13 Compute the determinants of: $\begin{bmatrix} 5 & \sqrt{3} + i \\ \sqrt{3} - i & 4 \end{bmatrix}$, $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 2 & 3 \\ 3 & 3 & 3 \end{bmatrix}$, and $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix}$

Solution. For the first matrix we have

$$\det \begin{bmatrix} 5 & \sqrt{3}+i \\ \sqrt{3}-i & 4 \end{bmatrix} = 5 \cdot 4 - (\sqrt{3}-1)(\sqrt{3}+1) = 20 - (3 + i\sqrt{3} - i\sqrt{3} - i^2) = 16$$

For the second matrix we have the following:

$$\begin{aligned} \det \begin{bmatrix} 1 & 2 & 3 \\ 2 & 2 & 3 \\ 3 & 3 & 3 \end{bmatrix} &= 1 \det \begin{bmatrix} 2 & 3 \\ 3 & 3 \end{bmatrix} + 2 \det \begin{bmatrix} 2 & 3 \\ 3 & 3 \end{bmatrix} + 3 \det \begin{bmatrix} 2 & 2 \\ 3 & 3 \end{bmatrix} \\ &= 1(6-9) + 2(6-9) + 3(6-6) \\ &= -9 \end{aligned}$$

For the third matrix we have the following:

$$\begin{aligned} \det \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix} &= \det \begin{bmatrix} 2 & 2 \\ 2 & 3 \end{bmatrix} + \det \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix} + \det \begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix} \\ &= 2 + 1 + 0 \\ &= 3 \end{aligned}$$

□

P.D.5 Use mathematical induction to prove that

$$1 + z + z^2 + \cdots + z^{n-1} = \frac{1 - z^n}{1 - z}$$

Proof. Let's begin with the case where $n = 1$. Then we have

$$1 = \frac{1 - z}{1 - z} = 1.$$

So we assume that the equation holds for n and we want to show that from there $n + 1$ also holds. Begin by setting $p_n(z) = 1 + z + z_2 + \cdots + z^{n-1}$. So we do the following work:

$$\begin{aligned}
 P_n(z) &= \frac{1 - z^n}{1 - z} \\
 P_n(z) &= \frac{1 - z^{n+1} - z^n + z_{n+1}}{1 - z} \\
 P_n(z) &= \frac{1 - z^{n+1}}{1 - z} - \frac{z^n(1 - z)}{1 - z} \\
 P_n(z) &= \frac{1 - z^{n+1}}{1 - z} - z^n \\
 P_n(z) + z^n &= \frac{1 + z^{n+1}}{1 - z} \\
 P_n(z) + z^n &= \frac{1 + z^{n+1}}{1 - z} \\
 1 + z + z_2 + \cdots + z^{n-1} + z^{n+1-1} &= \frac{1 + z^{n+1}}{1 - z}
 \end{aligned}$$

And we are done. □